



Clinical Practice Guideline

Sepsis Management

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical and Nursing

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Purpose

This document should be used in conjunction with the SCGOPHCG Adult Sepsis Pathway, which serves as a guide to clinicians dealing with patients who may have sepsis or septic shock. It is not a comprehensive guide to diagnosis or treatment of sepsis. It is intended to provide evidence-based treatment in an acute situation in a timely manner. This guideline does not replace clinical judgement and all clinicians using the Sepsis Pathway should seek senior clinical input in the case of any concerns about a particular patient.

Introduction

Sepsis is a time-critical medical emergency that arises when the body's response to an infection damages its own tissues and organs leading to organ failure. If not recognised promptly and treated, sepsis results in avoidable death and disability. Sepsis incidence is considered to have traditionally been underreported.

The Adult Sepsis Pathway, designed to meet the Clinical Care Standard, is divided into two parts, a Sepsis Trigger Tool followed by a Management Pathway.

- a. Many international, state-wide, and local Sepsis Trigger Tools have been designed to optimise sensitivity and specificity. A perfect tool will pre-emptively capture all those at risk of developing septic shock, whilst minimising over-escalation in those patients with an adequate immune response. The current screening tool uses early senior medical review, and the use of lactate, to optimise this balance.
- b. The management component clearly identifies investigations that need to be prioritised by medical and nursing staff within the first hour of care. These include oxygenation, investigation, antibiotic administration, source control, fluid resuscitation and vasopressor support, monitoring including lactate and senior medical review. Each component prompts clear documentation of what was done, when and by who.

Definitions

Sepsis	Defined as 'life-threatening organ dysfunction caused by a dysregulated host response to infection'. ¹
Septic Shock	Defined as 'subset of sepsis where underlying circulatory and cellular/metabolic abnormalities are profound enough to substantially increase mortality' requiring vasopressors to maintain a mean arterial pressure (MAP) of 65mmHg or greater AND serum lactate greater than 2mmol/L despite fluid resuscitation. ¹
Organ dysfunction	Organ dysfunction due to infection is identified by an increase in the Sequential [sepsis related] Organ Failure Assessment (SOFA) score of 2 points or more.

1. Sepsis Screening

Infection risk factors

Factors which predispose a patient to infection have been identified from large-scale epidemiological studies. These include age, vulnerable populations, impaired immunity, recent trauma or surgery and indwelling devices.

Risk factors/features of possible infection

- High risk physiological parameters in the presence of infection may be associated with organ dysfunction and hypoperfusion and should trigger the activation of the sepsis pathway
- The presence of any of these risk factors or features should prompt a screening for organ dysfunction and hypoperfusion; and active consideration of whether the patient could be septic. It is important to note that absence of risk factors or features of possible infections does not exclude sepsis.
- Moderate high-risk signs are based on a point-based score adapted from NICE UK (2016). Whilst the Sepsis Trigger Tool defaults to a lower threshold to activate the Sepsis Pathway, some critical care areas like ED, ICU, HDU and recovery have access to senior medical review at the screening stage. If available, these clinicians can end the Adult Sepsis Pathway early to avoid unnecessary investigation and the risk of false positive results.
- It is important to note that absence of risk factors or features of possible infections does not exclude sepsis. However, when sepsis has been deemed unlikely, another cause for the patient's deterioration must be sought.
- The use of Lactate is useful to identify organ dysfunction and hypoperfusion. Whilst a Lactate >2 in the presence of moderate-high risk signs should activate the Sepsis Pathway, Lactates ≤ 2 do not exclude sepsis.

2. Investigating Symptoms

The treatment of sepsis is time critical with roles clearly allocated to the medical and nursing team within the first hour.

- Investigations should focus on identifying the source, organ dysfunction and severity of disease.
- Blood culture remains the 'gold standard' for the detection of microbial pathogens related to bacteraemia and sepsis. Two (2) sets of blood cultures from separate sites (one from a central or peripheral line if present) should be taken before antibiotics are administered but should not delay the administration of antibiotics.
- Clearly label the BC samples as per Hospital Policy #48 Patient Identification and Procedure Matching – Communicating for Safety. Accuracy of the blood culture sample relies on drawing min 8- max 10mL into each bottle and minimising the risk of contamination.
- Investigation for other sources may include urine, sputum or wound cultures when applicable.
- Imaging by way of US, X-ray or CT should be guided by clinical presentation.

Source control

Additional strategies, besides antibiotics, may be required in the management of sepsis. Consider surgical causes or post-op cases requiring wash-out or drainage. Look for indwelling devices; lines ports or catheters and consider removal or replacement.

Maintain oxygen saturation

- Ensure that the airway is patent, and ventilation is adequate.
- Oxygen therapy should be administered in all septic patients with hypoxia in line with the Oxygen Therapy Clinical Practice Guideline.
- Oxygen administration should be commenced in patients with a blood gas analysis or with oxygen saturations less than 94%.
- If a patient is known to have, or is suspected of having chronic hypercarbia, and possible hypoxia- dependent respiratory drive, senior respiratory advice should be obtained.
- Doctors must document the target SpO₂, mode of oxygen administration and flow rate/
- Nurses must seek senior medical review in response to deteriorating oxygen saturations in patients with sepsis.

3. Treating Sepsis**Administer antimicrobial therapy**

- Use local guidelines (e.g., Neutropenic Sepsis Guideline), the eTG or in complex cases seek advice from Infectious Diseases to guide the early administration of antibiotics for suspected infection.
- One or more broad-spectrum intravenous antimicrobials should be started within 60 minutes from when sepsis is suspected. Do not delay administration of antibiotics because of the need to obtain blood culture results.
- The doctor should clearly document the choice of antibiotic/s made and the time the prescriptions was completed
- Make sure to alert the treating nurse that the antibiotics have been prescribed and need to be administered urgently
- Commence intravenous antibiotics after referring to the [Adult Sepsis Antibiotic Administration Table](#) and record the time the antibiotics were commenced
- As part of antimicrobial stewardship, once the pathogen is identified and sensitivities are established, empiric antimicrobial therapy should be narrowed or discontinued if the patient does not have an infection

Fluid resuscitation and vasopressors

- Doctors should prescribe IV isotonic crystalloid boluses up to a max of 30mL/kg for hypotension and/or lactate ≥ 2 and titrate to the clinical response
- The doctor should inform the treating nurse when fluids have been charted
- Nurses should document whether fluid was given, how much was given over the

hour and when it was started.

- Fluid resuscitation should be guided by the patients MAP, organ perfusion and their surrogate endpoints including urine output, mental state and lactate level.
- Senior medical review including ICU support should be sought in patients with persistent hypotension or features of hypoperfusion despite initial fluid boluses
- Consider vasopressor therapy if inadequate perfusion persists despite optimum fluid resuscitation.

4. Monitoring

- Septic patients need a higher level of vigilance to be able to recognise and respond to deterioration.
- Septic shock may appear suddenly and insidiously. Record and monitor observations at least hourly until the patient stabilises.
- Commence the Neurological Examination Chart and Fluid balance Chart.
- Evaluate the need for advanced measures such as vasopressors support, respiratory support, central venous access, arterial line and urinary catheter insertion.
- Lactate can be a useful marker to monitor organ hypoperfusion, but its interpretation may require senior input, especially when its trend does not align with that of the patient.
- Repeat the lactate level after 1hr, especially in the early management of sepsis or when the initial result is high
- Use lactate to help guide further fluid resuscitation or vasopressor management
- The nurse should document when the second lactate level was taken, and at what time senior input was requested especially if the lactate has not normalised or has not clearly trended towards normal
- Senior medical review is an integral step in the management of sepsis and should be requested at 1 and 4hrs after initiation of the Sepsis Pathway.
- If the senior nurse or JMO are concerned, and the supervising registrar is unavailable or delayed, the deterioration should be escalated to the consultant level

5. Documentation

- The Adult Sepsis Pathway does not replace more detailed documentation in the medical record
- The treating Medical Officer and Nurse, completing the pathway should document their designation and the date and time of completion

Compliance, monitoring and evaluation

The RRAD Committee will ensure that the following compliance, monitoring and evaluation is undertaken:

- Proportion of patients with suspected sepsis who had blood lactate levels taken as a part of screening for sepsis. (1a)
- assess adherence to the pathway and its performance (2b)
- Proportion of patients with sepsis who had blood cultures taken prior to starting antimicrobials. (3a)
- Proportion of patients with sepsis who had a diagnosis of sepsis recorded in their discharge summary. (6a)
- Proportion of patients with sepsis who had an unplanned readmission for any cause to any hospital within 30 days of discharge. (7a)
- Proportion of patients with sepsis who died during their inpatient episode of care. (8a)

Related Documents

1. Epidemiology of Sepsis in Australian Public Hospitals: A Mixed Methods, National Longitudinal Study (2013-2018)
https://www.safetyandquality.gov.au/sites/default/files/2020-05/epidemiology_of_sepsis_-_february_2020_002.pdf
2. Sepsis Clinical Care Standard
<https://www.safetyandquality.gov.au/standards/clinical-care-standards/sepsis-clinical-care-standard>
3. Review of trigger tools to support the early identification of sepsis in healthcare settings
https://www.safetyandquality.gov.au/sites/default/files/2021-05/review_of_trigger_tools_to_support_the_early_identification_of_sepsis_in_healthcare_settings_2.pdf
4. Sepsis Clinical Care Standard – Sepsis coordination roles and responsibilities
https://www.safetyandquality.gov.au/sites/default/files/2022-06/sepsis_coordination_roles_and_responsibilities_-_sepsis_clinical_care_standard.pdf
5. Sepsis: Risk stratification tools
[algorithm-for-managing-suspected-sepsis-in-adults-and-young-people-aged-18-years-and-over-in-an-acute-hospital-setting-2551485715 \(nice.org.uk\)](https://www.nice.org.uk/guidance/ng51/resources/algorithm-for-managing-suspected-sepsis-in-adults-and-young-people-aged-18-years-and-over-in-an-acute-hospital-setting-2551485715)
6. Neutropenic Sepsis
https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/SCGH/SCGOPHCG.CPG_Neutropenic_sepsis.pdf
7. Therapeutic Guidelines, Antibiotic
<https://tgldcdp-tg-org-au.gelibresources.health.wa.gov.au/topicTeaser?guidelinePage=Antibiotic&etgAccess=true>
8. Adult Sepsis Antibiotic Administration Table
https://www.cec.health.nsw.gov.au/data/assets/pdf_file/0007/596653/CEC-Adult-Sepsis-Antibiotic-Administration-Table.PDF

References

1. Sepsis Six by Ron Daniels UK Sepsis Trust: <http://dx.doi.org/10.1136/emj.2010.095067>
2. Sepsis: Recognition, diagnosis and early management (NICE Guideline 13/7/2016):
<https://www.nice.org.uk/guidance/ng51>
3. Sepsis: Risk stratification tools (NICE 2017):
<https://www.nice.org.uk/guidance/ng51/resources/algorithm-for-managing-suspected-sepsis-in-adults-and-young-people-aged-18-years-and-over-in-an-acute-hospital-setting-2551485715>

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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